English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

English transcription of the handwritten solutions. Calculations and unfinished answers are retained as written in the original.

1. Waiting for three heads

1. $$\mathbb{P}(Y=n)=\binom{n-1}{2}p^3(1-p)^{n-3}.$$

   There are two heads among the first $n-1$ tosses, and the nth toss is heads. Write $Y=X_1+X_2+X_3$, where the X_i are iid $\mathrm{Ge}(p)$; thus $\boxed{\mathbb{E}Y=3/p}$.

2. $$\mathbb{P}(Y\le 2k)=\sum_{i=3}^{2k}\binom{i-1}{2}p^3(1-p)^{i-3}.$$

   This can also be evaluated explicitly, but there is no need. As $k\to\infty$, $\mathbb{P}(Y\le 2k)\to 1$. For any random variable X, $\mathbb{P}(X\le k)\to 1$ as $k\to\infty$.

2. The two boxes

1. Intuitively, 5.5 looks like the right answer. It should not matter whether we choose a random subset and then a random element of it, or directly choose a random element of the initial set. A formal argument can be given in several ways, for example by calculating that each number is selected with probability 1/10. This is almost identical to the first question of Problem 1.6 in Test 1, 01.12.2024.

2. There are five pairs summing to 11: $(1,10),(2,9),\dots$. The probability of choosing any one pair is 1/4, and of not choosing it is 3/4. Thus

   $$\mathbb{P}(\text{none of the pairs is selected})=(\tfrac{3}{4})^5,\qquad \text{answer: } 1-(\tfrac{3}{4})^5.$$

3. If K is the number of slips transferred from A, then N is geometric conditionally on K. K is random, so an unconditional statement with a random parameter is not meaningful. The notes give the following conditional-expectation calculation:

   $$\mathbb{E}[N|K]=\frac{1}{K}\quad\Longrightarrow\quad \mathbb{E}N=\mathbb{E}[\mathbb{E}[N|K]]=\mathbb{E}[1/K],$$

   $$\mathbb{E}N=\sum_{n=1}^{10}\frac{1}{n}\mathbb{P}(K=n)=\frac{1}{10}(1+2+\cdots+10).$$

   The idea is the same as Problem 2 from Test 1 in Probability and Statistics, 2021.

3. Densities

The first three parts are standard. Recall

$$\mathbb{E}[\mathrm{Exp}(1)]=\int_0^\infty xe^{-x}\,dx=1,\qquad D[\mathrm{Exp}(1)]=1,\qquad \int_0^\infty x^2e^{-x}\,dx=2.$$

1. Normalization gives

   $$k\int_0^\infty xe^{-x}\,dx=1\quad\Longrightarrow\quad \boxed{k=1}.$$

   For the triangular joint-density region,

   $$\begin{aligned}1&=c\int_0^1\int_0^{1-x}(x+y)\,dy\,dx\\&=c\int_0^1\left((1-x)x+\frac{(1-x)^2}{2}\right)dx\end{aligned}$$

$$= \frac{c}{2}\int_0^1 (-x^2+1)\,dx = \frac{c}{2}(-\tfrac{1}{3}+1) = \frac{c}{3},$$

so $\boxed{c=3}$. Furthermore,

$$\mathbb{E}X = \int_0^\infty x^2 e^{-x}\,dx = 2.$$

By symmetry, $\mathbb{E}Y = \mathbb{E}Z$, and

$$\begin{aligned}
\mathbb{E}Y &= c\int_0^1\int_0^{1-x} y(x+y)\,dy\,dx \\
&= c\int_0^1 \left(x\frac{(1-x)^2}{2} + \frac{(1-x)^3}{3}\right)dx \\
&= \frac{c}{6}\int_0^1 (3x - 6x^2 + 3x^3 + 2 - 6x + 6x^2 - 2x^3)\,dx \\
&= \frac{c}{6}\int_0^1 (x^3 - 3x + 2)\,dx = \frac{c}{6}(\tfrac{1}{4} - \tfrac{3}{2} + 2) = \frac{1}{2}\cdot\frac{3}{4} = \frac{3}{8}.
\end{aligned}$$

2.

$$\mathbb{P}(X>2) = \int_2^\infty xe^{-x}\,dx, \qquad \int xe^{-x}\,dx = -\int x\,de^{-x} = -xe^{-x} - \int e^{-x}\,dx = -xe^{-x} - e^{-x}.$$

The handwritten numerical conclusion is $\mathbb{P}(X>2) = 0 + e^{-1} + e^{-1} = 2e^{-1}$. Also,

$$\begin{aligned}
\mathbb{P}(Y>\tfrac{1}{2}) &= c\int_0^{1/2}\int_{1/2}^{1-x} (x+y)\,dy\,dx \\
&= c\int_0^{1/2}\left[x(\tfrac{1}{2}-x) + \frac{(1-x)^2 - \frac{1}{4}}{2}\right]dx \\
&= c\int_0^{1/2}\frac{1}{8}(4x - 8x^2 + 4 - 8x + 4x^2 - 1)\,dx \\
&= \frac{3}{8}\int_0^{1/2}(-4x^2 - 4x + 3)\,dx = \frac{3}{8}(-4\cdot\tfrac{1}{24} - 4\cdot\tfrac{1}{8} + \tfrac{3}{2}) = \frac{3}{8}\cdot\frac{5}{6} = \frac{5}{16}.
\end{aligned}$$

3.

$$\mathbb{E}[YZ] = c\int_0^1\int_0^{1-x} xy(x+y)\,dy\,dx = \cdots = \frac{1}{10}.$$

Hence

$$\mathrm{Cov}(Y,Z) = \frac{1}{10} - \frac{9}{64} = -\frac{13}{320} \neq 0,$$

so Y, Z are not independent. For $\mathbb{E}[Z|Y = 3/8]$, use the conditional density. Since X and Y are independent, $\mathbb{E}[X|Y] = \mathbb{E}X$. The conditional expectation $\mathbb{E}[Z|Y]$ is the random variable $g(Y)$, where $g(y) = \mathbb{E}[Z|Y=y]$ is obtained as in the first case.

4. To find the density of YZ, put $V = YZ$, $W = Y$. The inverse is $Y = W$, $Z = V/W$, and

$$|J| = \begin{vmatrix} 0 & 1 \\ 1/w & * \end{vmatrix} = \frac{1}{w}.$$

Thus

$$f_{V,W}(v,w) = f_{Y,Z}(w, v/w)\frac{1}{w} = c\left(w + \frac{v}{w}\right)\mathbb{1}_{\{w,v>0,\ w+v/w<1\}}\frac{1}{w}.$$

Since $w>0$, $w + v/w < 1$ is equivalent to $w^2 - w + v < 0$. The roots are

$$w_{1,2} = \frac{1 \pm \sqrt{1-4v}}{2}.$$

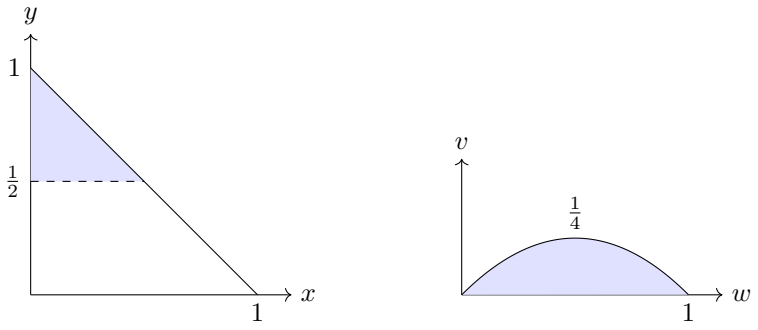
The region lies below $v = w - w^2$, so $0 < v < 1/4$ and

$$\frac{1-\sqrt{1-4v}}{2} < w < \frac{1+\sqrt{1-4v}}{2}.$$

Consequently,

$$f_V(v) = 3\int_{(1-\sqrt{1-4v})/2}^{(1+\sqrt{1-4v})/2}\left(1 + \frac{v}{w^2}\right)dw$$

$$
\begin{aligned}
&= 3\left[\sqrt{1-4v} - \frac{2v}{1+\sqrt{1-4v}} + \frac{2v}{1-\sqrt{1-4v}}\right] \\
&= 3\left[\sqrt{1-4v} - \frac{2v(1-\sqrt{1-4v})}{4v} + \frac{2v(1+\sqrt{1-4v})}{4v}\right] \\
&= 6\sqrt{1-4v}, \qquad 0 < v < 1/4.
\end{aligned}
$$



4. Random permutations

1. Choose the other $k-1$ elements in the cycle containing 1, arrange them, and permute the remaining elements:
$$\mathbb{P}(1 \text{ is in a cycle of length } k) = \frac{\binom{n-1}{k-1}(k-1)!(n-k)!}{n!} = \frac{(n-1)!}{n!} = \frac{1}{n}.$$

2. Observe
$$\sum_{i=1}^{n} \mathbb{1}_{\{i \text{ is in a cycle of length } k\}} = k\,(\text{number of cycles of length } k).$$
Taking expectations and using part 1 gives $n/n = k\,\mathbb{E}[\text{number of cycles of length } k]$. By linearity,
$$\mathbb{E}[c(\sigma)] = \mathbb{E}\left[\sum_{k=1}^{n} \text{number of cycles of length } k\right] = 1 + \frac{1}{2} + \cdots + \frac{1}{n}.$$
For fixed points,
$$\mathbb{P}(\sigma(1) = 1) = \mathbb{P}(1 \text{ is fixed}) = \mathbb{P}(i \text{ is fixed}) = \mathbb{E}[\mathbb{1}_{\{i \text{ is fixed}\}}] = \frac{1}{n}.$$
Thus $\mathbb{E}[F(\sigma)] = \mathbb{E}[\sum_{i=1}^{n} \mathbb{1}_{\{i \text{ is fixed}\}}] = n/n = 1$.

3. This is Problem 21 in the file on Moodle.

4. Yes; use an appropriate CLT for weakly dependent random variables. One can show that $\mathbb{E}[c(\sigma)]$ and $D[c(\sigma)]$ are asymptotic to $\log n$, and consider how to represent $c(\sigma)$ as a sum of random variables.